C-DE324 – Deliverable #1

Data Warehouse Project

Course: DE324: Big Data Engineering

Semester: Spring 2025

Submission Date: March 23, 2026

DB Source Name: Online Retail II (UCI ML Repo)

Business Area: Sales Analysis

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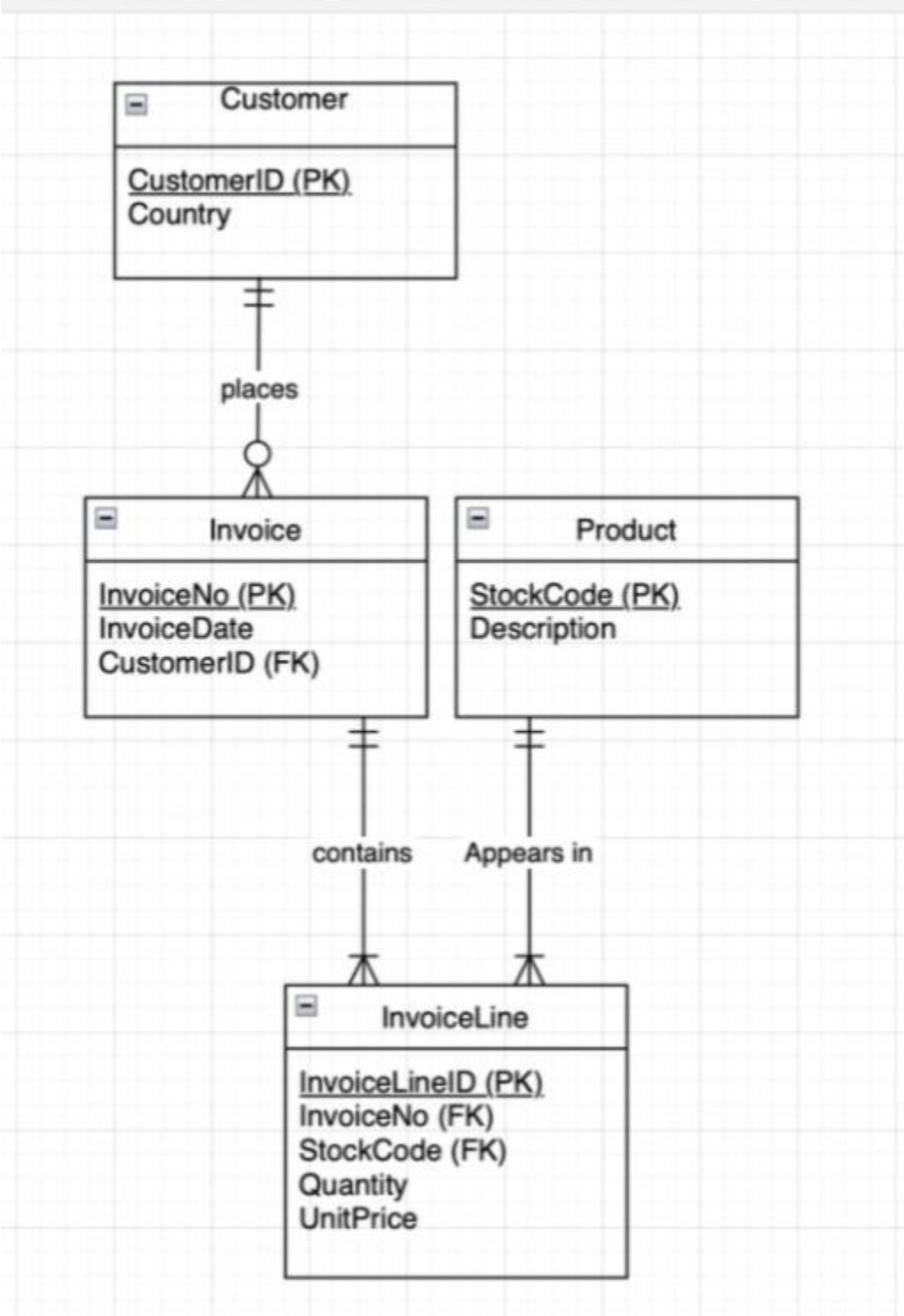
Business Goals

The goal of this data warehouse is to analyze sales performance to support decision-making.

Key questions that can be answered using the warehouse:

- What are the top-selling products?
- How do sales vary over time (daily, monthly, yearly)?
- Which customers or countries generate the most revenue?
- What is the average order value?
- How do quantities and revenue trends change across regions?

Source ERD


Customer
CustomerID (PK)
Country
places
Invoice
InvoiceNo (PK)
InvoiceDate
CustomerID (FK)
Product
StockCode (PK)
Description
contains
Appears in
InvoiceLine
InvoiceLineID (PK)
InvoiceNo (FK)
StockCode (FK)
Quantity
UnitPrice

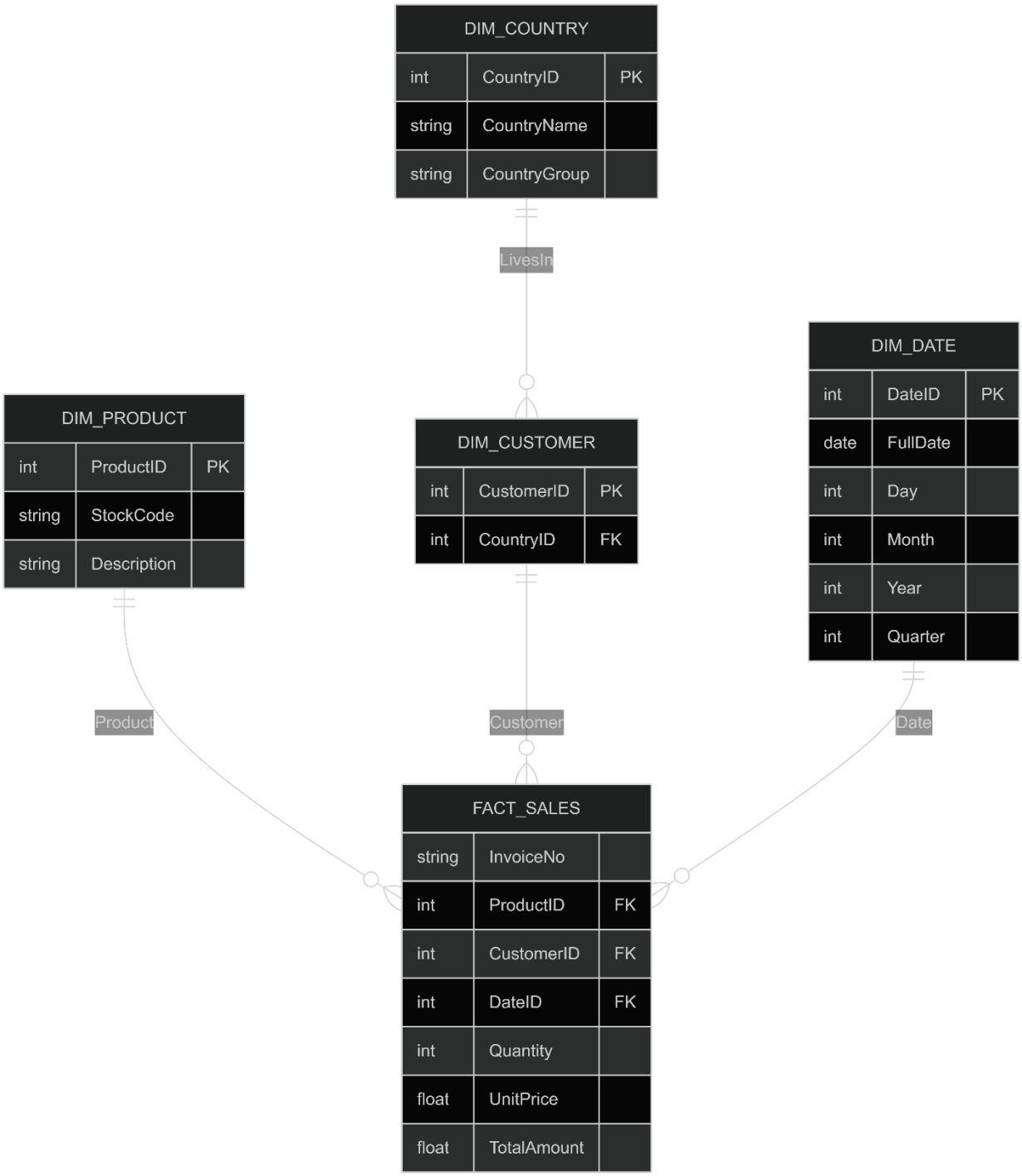
Tables:

- Customer
  - CustomerID (PK)
  - Country
- Invoice
  - InvoiceNo (PK)
  - InvoiceDate
  - CustomerID (FK)
- Product
  - StockCode (PK)
  - Description
- InvoiceLine
  - InvoiceLineID (PK)
  - InvoiceNo (FK)
  - StockCode (FK)
  - Quantity
  - UnitPrice

Relationships:

- Customer places Invoice (1:M)
- Invoice contains InvoiceLine (1:M)
- Product appears in InvoiceLine (1:M)

Star Schema


DIM_COUNTRY
int CountryID PK
string CountryName
string CountryGroup
LivesIn
DIM_DATE
int DateID PK
date FullDate
int Day
int Month
int Year
int Quarter
DIM_PRODUCT
int ProductID PK
string StockCode
string Description
DIM_CUSTOMER
int CustomerID PK
int CountryID FK
Product
Customer
Date
FACT_SALES
string InvoiceNo
int ProductID FK
int CustomerID FK
int DateID FK
int Quantity
float UnitPrice
float TotalAmount

Fact Table: FactSales

Grain: one row per product per invoice line

Columns:

- InvoiceNo (Degenerate Dimension)
- ProductID (FK)
- CustomerID (FK)
- DateID (FK)
- Quantity (measure)
- UnitPrice (measure)
- TotalAmount = Quantity × UnitPrice (derived measure)

Dimensions

1. DimDate

- DateID (PK)
- FullDate
- Day
- Month
- Year
- Quarter

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2. DimCustomer

- CustomerID (PK)
- CountryID (FK)

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3. DimProduct

- ProductID (PK)
- StockCode

- Description

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4. DimCountry

- CountryID (PK)
- CountryName
- CountryGroup

Schema Description

- Fact Table

FactSales

- Stores measurable business events (sales transactions)
- Linked to all dimensions using foreign keys

- Measures
  - Quantity: number of items sold
  - UnitPrice: price per item
  - TotalAmount: total revenue (derived)

- Dimensions
  - Date → time analysis
  - Customer → customer-based insights
  - Product → product performance
  - Country → geographic analysis